

## Experiment 6

### Explainability for Image data using XRAI

**Aim:** To understand the concept of Explainable AI (XAI) and implement the XRAI (eXplanation with Ranked Area Integrals) attribution method to visually interpret the predictions of a deep Convolutional Neural Network (CNN) image classifier.

#### Stages for XRAI:

1. **Over-segmentation:** The input image is over-segmented into many small, coherent regions ("superpixels") using Felzenszwalb's graph-based segmentation algorithm, typically at multiple granularities.
2. **Pixel-level attribution:** Integrated Gradients is computed for every pixel with respect to the target class, usually averaged over multiple random baselines (e.g. black and white images) to reduce noise.
3. **Region aggregation:** The pixel-level IG scores are summed within each segment to obtain a per-region importance score.
4. **Greedy ranking:** Regions are greedily ordered by importance — at each step, the region that adds the most incremental attribution to the already-selected area is chosen next.
5. **Heatmap construction:** The final XRAI map assigns higher intensity to regions selected earlier in the ranking, so the top 10-30% of the map highlights the most decision-relevant object parts.

#### Procedure:

1. Install the required libraries: tensorflow, saliency, scikit-image, matplotlib.
2. Load the pretrained InceptionV3 model with ImageNet weights.
3. Load a sample RGB image and resize it to 299×299 (InceptionV3's expected input size).
4. Run a forward pass to obtain the model's predicted class and confidence.
5. Define a `call_model_function` that returns input-output gradients for the target class — this is the interface the saliency library needs.
6. Instantiate `saliency.XRAI()` and call `GetMask()` to compute the XRAI attribution map.
7. Visualize: (a) the original image, (b) the raw XRAI heatmap, (c) the image masked to show only the top-ranked (most influential) regions.
8. Repeat for 3-5 different images/classes and compare which regions the model relies on.

Task 1: Before computing XRAI, predict which region the model will rely on, and one region you expect it to ignore. The guess should be recorded in the notebook and compared against the actual XRAI heatmap once revealed.

Task 2: Using interactive sliders, drag a gray patch across the image and watch the model's confidence score update live. The challenge: find the smallest patch that causes the largest confidence drop, and check whether it lines up with the region XRAI highlighted.